Hybrid search and integer repair for the prize-collecting covering tour problem

Working manuscript

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Abstract

Research manuscript in preparation. Scientific claims, numerical results and references will be included only after verification.

1 Introduction

Selecting locations and ordering their visits are coupled decisions in the prize-collecting covering tour problem (PCCTP). A tour must include mandatory locations, serve demands through nearby visited locations and collect a minimum prize. An optional location can contribute to both coverage and prize collection, while also changing the routing cost. Consequently, a shorter ordering of a fixed visited set and a better choice of that set are different sources of improvement. An algorithm that changes visits must account for the feasibility requirements that those visits were satisfying.

Existing PCCTP methods already address this interaction through several optimization components. Reactive GRASP with GENIUS construction and variable neighborhood descent provides a heuristic precedent [2]. Reduction rules, valid inequalities, branch-and-cut and a heuristic–integer programming hybrid have also been studied [3]. The later MDM-GRASP approach uses patterns mined from high-quality solutions [4]. These precedents make a comparison with construction alone insufficient and prevent treating generic hybridization as a contribution by itself. The relevant question is whether a specified combination improves the solution process after its additional work is charged.

This study evaluates destruction and repair after MDM initialization, with route ordering performed by LKH and with greedy and restricted integer-repair alternatives examined separately. It asks three questions. Does changing the visited set add value beyond reordering the same set from common starts? Does joint integer insertion improve on greedy repair after matched destructions? Finally, does a complete hybrid improve on a sufficiently developed MDM reference under an explicit total-time comparison? Separating these questions avoids attributing a strong ordering component's gains to repair, or presenting a conditional improvement from an existing solution as a complete algorithmic advantage.

The completed evidence is exploratory. It includes a positive short-budget training comparison, a failed validation screen, a conditional fixed-support replay and an unfavorable mean result for the evaluated joint-insertion MIP variant. The manuscript reports these outcomes with their denominators, geometry-level sensitivity and independent feasibility checks. The longer-horizon comparison and the iteration-limited reference study

remain necessary before making a broader performance claim. The formulation, neighborhood descriptions and audited exploratory results presented here therefore do not establish a new best-known solution, a novel general hybridization principle or superiority over the published PCCTP methods.

2 Related work

PCCTP heuristics already combine construction, local improvement and diversification. Clímaco et al. [2] developed a reactive GRASP using an adapted GENIUS construction and randomized variable neighborhood descent. Their local search changes both route order and visited locations while respecting the problem's feasibility requirements. This provides a stronger point of comparison than a construction heuristic alone. The experimental question here concerns the additional value of a separate destruction-and-repair phase after initialization, rather than the first use of multiple neighborhoods for the PCCTP.

Exact optimization and mathematical-programming hybrids also have direct precedents for this problem. Clímaco et al. [3] presented a branch-and-cut algorithm, valid inequalities, reduction rules and a hybrid combining a PCCTP heuristic with integer programming. Consequently, neither adding preprocessing nor combining a GRASP-derived method with a MIP solver is sufficient by itself to distinguish the present investigation. Any useful extension must be specified more precisely and evaluated against appropriate controls, including its model-building and optimization time.

The later data-mining approach of Clímaco et al. [4] exploits patterns in high-quality solutions to guide GRASP search. Its publisher description includes analyses of pattern mining, statistical comparisons and time-to-target behavior. The supplied MDM code and manuscript support the recovered reference used here. We distinguish that reference's declared corrections and parameter choices from the published method description. In particular, a comparison ending before mining is activated cannot establish superiority over the mining mechanism. Final-cost comparisons and mining-exposure records address different parts of that question.

Destruction and repair also belong to the broader large-neighborhood-search literature. Ropke and Pisinger [8] used competing subheuristics with selection frequencies reflecting their historical performance in adaptive large neighborhood search for pickup and delivery with time windows. The uniform-removal configuration evaluated here has no such adaptive operator-selection layer. It should therefore not be described as an implementation or improvement of that complete framework. Its temporary coverage and prize violations, greedy reinsertion and acceptance rule define the specific trajectory whose incremental value is examined.

The route-ordering component creates another attribution problem. Helsgaun [7] developed an effective implementation of the Lin–Kernighan traveling-salesman heuristic. In the present implementation, LKH receives a symmetric routing matrix for a fixed set of visited locations. Reordering that set cannot repair a missing coverage location or add prize; those effects require visit changes. Conversely, an observed hybrid gain could arise from strong reordering rather than the visit-change mechanism. The fixed-support component replay is designed to separate these effects, while retaining its conditional status and charged remaining allowance.

General-purpose MIP neighborhood methods offer related alternatives. Fischetti and Lodi [6] used local branching cuts to define solution subspaces explored by a MIP solver.

Danna, Rothberg and Le Pape [5] introduced relaxation-induced neighborhood search, which uses information from a continuous relaxation and formulates neighborhood exploration as a further MIP. These methods motivate exact exploration around feasible incumbents, but they do not identify a unique PCCTP neighborhood. Fixing routing edges and visit decisions, contracting retained paths, and restricting insertion positions impose distinct structural choices. Their feasibility, computational cost and improvement potential require separate evaluation; a local optimality certificate is limited to the decisions left free.

This investigation therefore separates three empirical questions: whether changing visits adds value beyond fixed-support reordering, whether restricted integer repair adds value beyond greedy repair on matched destructions, and whether any resulting complete search improves on a sufficiently developed MDM reference under an explicit total-time comparison. The completed exploratory tests include unfavorable outcomes and a failed validation screen. The ongoing longer-horizon work does not yet settle the third question. Differences in neighborhood construction are descriptions of the method under study, not evidence of novelty or of a literature-wide performance advantage.

3 Problem definition and mathematical formulation

Let $V = T \dot{\cup} R$ be the set of locations that can be visited, where T contains the mandatory locations and R contains the optional ones. Let W be a disjoint set of demand locations. The routing graph is the complete undirected graph $G = (V, E)$, with $E = \{\{i, j\} : i, j \in V,\ i < j\}$ and edge costs c_e. For every $w \in W$, let $N_w \subseteq V$ denote its admissible covering locations. In the supplied instances, N_w consists of the visitable locations whose input distance to w does not exceed the coverage threshold. A nonnegative prize p_i is collected when $i \in V$ is visited, and the required total prize is P. A demand location is served when at least one location in N_w is visited. Coverage itself does not add a prize to the visited locations' total.

The prize-collecting covering tour problem seeks a minimum-cost simple cycle that visits every location in T, covers every demand in W, and collects at least P units of prize. This is the problem addressed by the supplied MDM-GRASP implementation and manuscript [4]. We assume $T \neq \varnothing$ and select a root $r \in T$, consistently with the benchmark instances used here. No triangle inequality is needed for the formulation below.

For $S \subseteq V$, let $\delta(S)$ be the set of edges with exactly one endpoint in S, and write $x(F) = \sum_{e \in F} x_e$ for $F \subseteq E$. Binary variables y_i indicate visited locations, and binary variables x_e indicate used routing edges. The following degree and connectivity formulation

serves as a reference model:

$$\min \quad \sum_{e \in E} c_e x_e \tag{1}$$

$$\text{subject to} \quad x(\delta(\{i\})) = 2y_i \qquad i \in V, \tag{2}$$

$$y_i = 1 \qquad i \in T, \tag{3}$$

$$\sum_{i \in N_w} y_i \geq 1 \qquad w \in W, \tag{4}$$

$$\sum_{i \in V} p_i y_i \geq P, \tag{5}$$

$$x(\delta(S)) \geq 2y_i \qquad \varnothing \neq S \subseteq V \setminus \{r\},\ i \in S, \tag{6}$$

$$\sum_{i \in V} y_i \geq 3, \tag{7}$$

$$x_e \in \{0,1\} \qquad e \in E, \tag{8}$$

$$y_i \in \{0,1\} \qquad i \in V. \tag{9}$$

Constraints (2) assign degree two to visited locations and degree zero to unvisited ones. Constraints (4) and (5) enforce coverage and prize collection, respectively. The rooted connectivity inequalities (6) exclude selected components disconnected from r. Constraint (7) explicitly states the simple-cycle convention; for integer solutions it also follows from the degree equations, binary edges, and the mandatory root. The objective minimizes the total cost of the selected routing edges. This reference formulation is not claimed as a new formulation.

To verify the integer interpretation, any feasible PCCTP cycle directly satisfies the model. Conversely, the degree equations imply that the graph induced by selected edges is a collection of vertex-disjoint simple cycles. If a selected component does not contain r, choosing its vertex set as S in (6) gives a contradiction. Thus all selected locations belong to one cycle containing the mandatory root. The remaining constraints give the required visits, coverage, and prize total.

Fixing a subset of these routing or visit decisions defines a restricted repair problem. An optimality certificate for that restricted problem is local to its fixed decisions. By itself, it does not establish global optimality when the fixed decisions exclude feasible solutions of the reference problem. This distinction applies to the contracted repair experiments reported in our research record.

4 Structured neighborhoods and conditional feasibility

A repair neighborhood is defined relative to a feasible incumbent $(\bar{x}, \bar{y})$. Let $U \subseteq R$ contain the optional locations whose visit decisions may change, and let $F \subseteq E$ contain the routing edges whose decisions may change. The repair problem consists of the reference formulation with the additional restrictions

$$x_e = \bar{x}_e \qquad e \in E \setminus F, \tag{10}$$

$$y_i = \bar{y}_i \qquad i \in V \setminus U. \tag{11}$$

The incumbent satisfies these restrictions and therefore provides a feasible fallback. Optional locations outside U keep their incumbent visit status; mandatory locations remain selected through the reference formulation. The choice of F must be stated together with U: releasing a visit alone need not permit its removal when the required reconnection edges remain fixed.

This neighborhood separates the feasibility of the mathematical problem from the behavior of a time-limited implementation. The incumbent remains mathematically feasible even when a solver does not return a new solution. An implementation can preserve it explicitly and validate every returned cycle against the original visits, coverage sets, prize quota, and edge costs. A returned feasible solution gives a valid PCCTP upper bound. In contrast, a lower bound for the restricted repair problem need not be a lower bound for the unrestricted PCCTP, because the restrictions may exclude better tours.

4.1 Contraction of retained paths

Let H be the graph formed by the incumbent edges fixed to one in (10). Since these edges belong to a simple cycle, each nontrivial component of H is a path, unless the entire incumbent cycle is retained. Every retained path must occur in every feasible repair. Its interior vertices already have degree two, while each endpoint requires one additional incident edge. An isolated selected vertex instead requires two incident edges. An isolated released vertex may be selected or omitted.

The path components can consequently be represented by contracted nodes with their endpoint incidences retained explicitly. All routing edges fixed to zero remain unavailable. A retained internal edge is not made available again merely because both of its endpoints appear in a boundary set. The cost of fixed edges is collected in the constant

$$c_{\text{fixed}} = \sum_{e \in E \setminus F \colon \bar{x}_e = 1} c_e. \tag{12}$$

Prizes and demand coverage of retained path vertices are accounted for in the corresponding mandatory component. Degree requirements at the original endpoints and connectivity between selected components complete the contracted model. The endpoint representation is needed because an ordinary degree-two constraint on a contracted node would not distinguish which original endpoint an edge uses.

The integer correspondence follows by contracting the fixed paths of a feasible repair tour. Conversely, a connected contracted solution satisfying the endpoint requirements expands uniquely along each retained path, up to its traversal direction. The expanded routing edges satisfy the original degree equations and all fixed decisions. Preserving the original prize and coverage contributions then yields a feasible repair tour with objective equal to the contracted edge cost plus c_{fixed}. If a retained component is already the incumbent cycle, its vertices have no remaining degree capacity. Connectivity precludes adding a separate cycle, so the incumbent routing cycle is the only feasible routing solution of that repair problem.

The contraction is used as a representation of a specified neighborhood, rather than as a claim that the neighborhood is new. Its correctness does not imply that a chosen neighborhood contains an improving solution, that a short solve will find one, or that contraction improves overall running time. Those are separate empirical questions. In particular, a short offline repair from an incumbent generated earlier is distinct from an end-to-end algorithm whose construction, selection, model assembly, and repair all share one time budget.

5 Hybrid search with coverage and prize repair

This section specifies an experimental hybrid search; it does not assert novelty of its components or superiority over published algorithms. Figure 1 shows the search flow and the position of the separately evaluated integer-repair alternative.

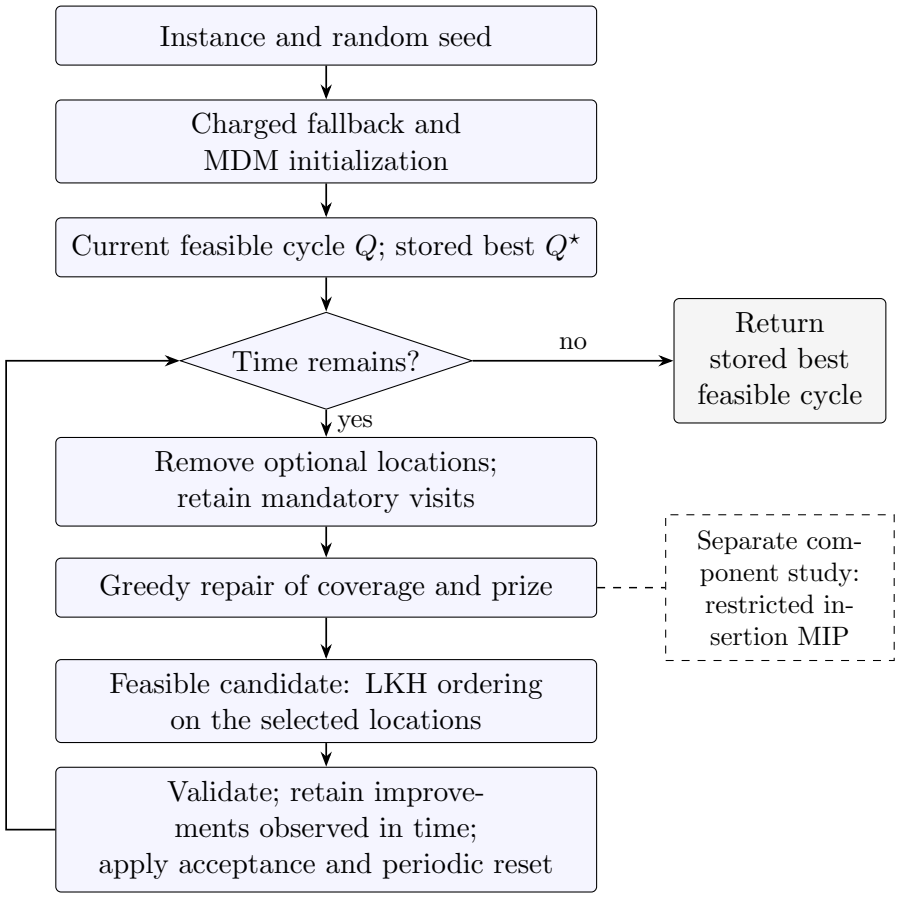


Figure 1: Search flow under one charged outer time budget. The solid path shows successful candidate generation. Search also stops if no optional location can be removed. An infeasible or interrupted repair skips ordering and preserves the stored feasible best. Only improvements observed by the deadline qualify, even if cleanup returns later. The dashed side box identifies the insertion-MIP component comparison; it is not an additional stage in the tested greedy-repair search.

5.1 Destruction and feasibility repair

Let Q be the current feasible cycle, $S(Q)$ its visited set, and $Q^\star$ the best accepted cycle. A random destruction removes $k = \min\{6, |S(Q) \cap R|, |S(Q)| - 3\}$ optional locations, sampled uniformly without replacement. Mandatory locations are retained. The trajectory stops if $k = 0$. A diagnostic alternative removes locations closest in input edge cost to a random optional seed; it is evaluated separately from uniform removal.

Destruction can violate coverage and prize collection. Figure 2 illustrates how inserting a different optional location can restore both requirements. For the remaining visited set

S, define

$$U(S) = \{w \in W : N_w \cap S = \varnothing\}, \qquad D(S) = \max\{0, P - \sum_{i \in S} p_i\}.$$

For an absent location v, the least insertion cost is

$$\Delta(v, Q) = \min_{(a,b) \in Q} \{c_{av} + c_{vb} - c_{ab}\},$$

where consecutive locations include the closing edge. With $\bar{p} = \max\{1, |V|^{-1} \sum_{i \in V} p_i\}$, define

$$g(v, S) = |\{w \in U(S) : v \in N_w\}| + \frac{\min\{p_v, D(S)\}}{\bar{p}}.$$

Among locations with positive gain, repair minimizes $\xi_v \Delta(v, Q)/g(v, S)$, with independent $\xi_v \sim \text{Uniform}[0.8, 1.2]$. It inserts that location in a least-cost position, updates coverage and prize, and repeats until $U(S) = \varnothing$ and $D(S) = 0$. Ties follow location and position indices. The random multiplier also applies to negative insertion costs; no triangle inequality is assumed. Repair is abandoned upon exhaustion of time or eligible locations, preserving the stored feasible incumbent.

Every returned repair is checked against the original instance. Removing only optional locations preserves mandatory visits, inserting only absent locations preserves distinctness, and the stopping conditions enforce coverage and prize. These properties justify feasibility of a completed repair; they provide no guarantee of an improving objective value.

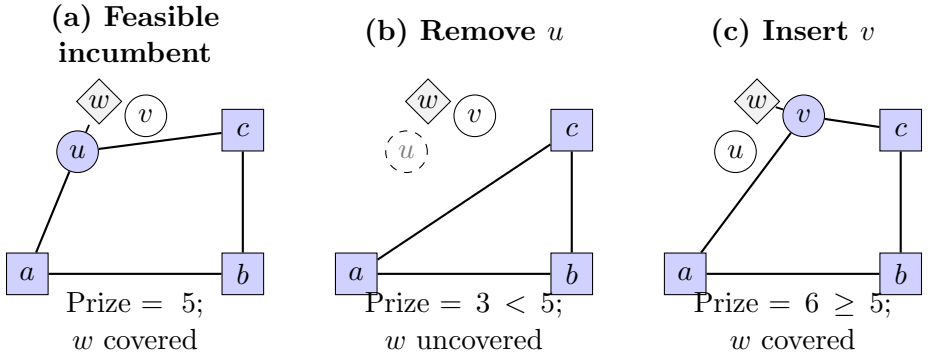


Figure 2: Illustrative destruction and repair, not a benchmark outcome. Squares are mandatory locations, circles are optional locations, and the diamond is a demand. Filled visit nodes belong to the cycle; solid lines are routing edges and dashed links show active coverage. Let $p_a = p_b = p_c = 1$, $p_u = 2$, $p_v = 3$, $P = 5$, and $N_w = \{u, v\}$. Removing u preserves three mandatory visits but violates both requirements. One feasible repair inserts v, restoring coverage and the prize quota. The drawing does not assert that every randomized repair selects v or that an improvement is guaranteed.

5.2 Ordering and acceptance

The repaired sequence is passed to LKH version 3.0.6 with an explicit symmetric cost matrix and the visited set fixed. Each invocation has at most 0.2 seconds, including matrix-file construction and subprocess execution. A validated output replaces the order

only when it improves cost. Otherwise, including after a timeout, the repaired input order remains available. Passive counters distinguish successful exits, subprocess timeouts, and calls without a successful exit. These measurements must be examined before attributing poor performance to the neighborhood rather than its implementation.

For a candidate Q', the best cycle is updated only by an improvement accepted before the outer deadline. The current cycle is replaced whenever $c(Q') \leq 1.01c(Q^\star)$, using the best cost after any accepted improvement. Every twentieth destruction attempt resets the current cycle to $Q^\star$. The current solution can therefore worsen temporarily while the accepted best cost is nonincreasing. Destruction size, acceptance threshold, and reset interval are frozen before a batch. This configuration has no learned policy or adaptive operator weighting.

5.3 Charged initialization and control

Both total-time arms first construct a nearest-neighbor cycle through all visitable locations and attempt beneficial feasible optional deletions under a soft construction cap of 0.25 seconds. Completing and validating the cycle can exceed that soft cap; construction and setup are charged to the outer budget, and the fallback qualifies only before that deadline. The supplied benchmark instances admit the all-visit fallback.

The control continues the recovered MDM-GRASP under a 60-second deadline. The hybrid reserves a 10-second MDM deadline and uses the remaining outer time for destruction and repair. Both native deadlines debit the setup and fallback time already spent. The recovered baseline uses the isolated decoder-index and occupied-elite-count corrections, which must be disclosed separately from algorithmic modifications. A predeadline native export seeds the hybrid when it improves the fallback; otherwise the fallback is retained. No expensive offline incumbent seeds this total-time comparison.

Native cleanup may return after its search deadline. A separate safety timeout limits the process but does not extend eligibility of solutions. Return-time overruns and the resulting loss of time for repair are retained in the records. The budget concerns accepted solution timestamps, not a guarantee that every operating-system process returns within exactly 60 seconds.

5.4 Deadline recording and component attribution

The prospective collector records absolute `CLOCK_MONOTONIC` timestamps for improvements and parent start time t_0. An event qualifies when its timestamp is at most $t_0 + 60$, even if routine cleanup finishes later. An earlier collector instead required the entire repair routine to return before the deadline. Its original outputs are preserved separately, and reconstructed event bounds are treated as sensitivity data rather than replacements for prospective measurements.

A conditional component control replays the exact saved pre-repair cycle under its recorded remaining allowance, holding its visited set fixed and allocating that allowance to LKH ordering with a fixed large trial cap. This asks whether changing visits contributes beyond reordering the same initial support. The control is a conditional replay, not a new independent end-to-end run. It requires original scalar cost and feasibility checks, unchanged support, source hashes, and deadline checks. This comparison must precede any attribution of a performance gain to coverage and prize repair.

6 Experimental protocol

6.1 Benchmark scope and experimental units

The development process separated geometries rather than individual routes or random seeds. The principal training screen uses nine difficult instance variants: a280, gil262, gr229, kroA200, pr226, rd400, si175, ts225 and tsp225. Each geometry contributes one fixed variant, whose mandatory, optional and demand sets and prize quota are retained unchanged across methods. Complete instance identifiers and input hashes are stored in the batch manifests. For these nine training files, the local SHA-256 digests match the corresponding public file-information records in the Mendeley version-2 deposit cited by the earlier PCCTP work [1]. This identity check covers the nine selected input variants; the remaining benchmark files and historical evaluation subsets are outside its verified scope. These selected cases are not a representative random sample of all PCCTP instances and do not reproduce the full published benchmark evaluation.

The completed 60-second validation screen uses seven additional geometries. Those geometries had already informed an earlier configuration and are therefore treated as reused validation data, rather than untouched confirmation. Twelve further test geometries remain unopened. Changing the time budget or random seeds does not restore independence to data previously used during method development. Results from selected training instances, reused validation and conditional component replays are reported separately.

An inspection of the nine training inputs also distinguishes coverage requirements from prize requirements. In gr229, kroA200, pr226 and rd400, the mandatory locations already cover every demand. Coverage constraints are consequently redundant in these four complete instances, although the prize quota and routing decisions remain restrictive. An improvement on these cases cannot be attributed to restoring uncovered demands. This structural observation follows directly from the inputs and does not identify the cause of any observed cost difference.

6.2 Baseline and computing environment

The reference implementation is the supplied MDM-GRASP associated with the data-mining study [4]. It is compiled separately from the experimental Python drivers. Two isolated corrections initialize an otherwise uninitialized decoder index and bound elite-bank access by the number of occupied entries. The legacy source is preserved. Results of the corrected implementation are identified accordingly; they are not described as measurements of an unmodified published executable.

The reference configuration uses 70 maximum iterations, construction parameter $\alpha = 0.6$, an elite-bank capacity of ten, absolute mining support two, and a stability threshold of five percent of the iteration limit. Absolute support two corresponds to 20 percent only when the bank contains ten solutions; the experiment does not silently recompute it for a partially occupied bank. The reference policy bypasses experimental learned rankings and adaptive timing decisions. Instrumentation and collection still impose overhead and are not assumed to be cost-free.

The local environment is an Apple M2 with 16 GiB of memory and eight physical cores. The recorded native compiler is GCC 15.1.0; the research driver uses Python 3.9, SciPy 1.13 and HiGHS through the available Python interfaces. The ordering component is LKH 3.0.6. Program and binary hashes identify the executables used by each batch.

Only one research optimizer is scheduled at a time. This scheduling constraint does not establish a constant background load, temperature or effective processor frequency. Raw seconds are not converted into speedups against the different processor used in a prior paper by scaling clock frequencies.

6.3 Completed experiments and longer-horizon design

The completed total-time training experiment allocates 60 seconds to each method, using nine geometries and three matched seeds. The completed validation screen uses seven geometries and five seeds. Both start from a fresh, charged fallback and include initialization within the outer budget. The fixed-support replay instead starts from the saved pre-repair cycle and uses its recorded remaining allowance. Generation of that shared state is excluded from the replay, which is therefore a conditional component test. The matched-destruction MIP experiment similarly shares saved MDM states; its comparisons must not be interpreted as complete algorithms with free incumbent generation.

The longer-horizon training design allocates 300 seconds to each of three methods: the reference, a hybrid with ten seconds of reference initialization, and a hybrid with 60 seconds of reference initialization. All use the same nine variants and seeds 70–72. Method order is shuffled reproducibly within each instance–seed block. Destruction, repair, ordering and acceptance parameters are frozen before collection. Both hybrid configurations are retained in the report, regardless of their relative performance. This design schedules 81 executions and remains an exploratory training screen. Its incomplete outcomes are excluded from the completed-results table.

A separate prospective design addresses the 70-iteration stopping criterion. An initial nine-case calibration with seed 80 must complete and pass its audits before the ten-seed comparison starts. In that subsequent comparison, each of the nine training variants uses seeds 81–90. A reference run first completes all 70 iterations and provides its measured process-wall duration $T_{g,s}$. Each hybrid then receives that same outer solution-eligibility horizon, including its own fallback, setup and initialization. Initialization is capped at $\min\{10, T_{g,s}\}$ or $\min\{60, T_{g,s}\}$, respectively. Neither the reference route nor its objective value is supplied as a hybrid starting solution. Independent post-exit auditing does not enlarge a search budget.

The resulting horizon is conditioned on the measured reference duration; it is not an independently fixed common deadline. The reference necessarily precedes the candidates in a pair. Shuffling pair order and the two candidate orders does not remove that chronological asymmetry, which is disclosed as a limitation. The fixed-300-second design supplies a complementary comparison with shuffled order among all three methods. The ten-seed design schedules 90 reference runs and 180 candidate runs if every reference completes. It is not yet a completed experiment and does not constitute a replication of the full published instance set.

6.4 Outcome aggregation and failure handling

For geometry g, let $\bar{c}_{B,g}$ and $\bar{c}_{H,g}$ be the arithmetic mean final costs over all its scheduled seeds for the reference and a candidate, respectively. The primary descriptive effect is

$$G_g = 100\frac{\bar{c}_{B,g} - \bar{c}_{H,g}}{\bar{c}_{B,g}}, \qquad \bar{G} = \frac{1}{|\mathcal{G}|}\sum_{g\in\mathcal{G}} G_g. \tag{13}$$

Positive values favor the candidate. This equally weighted geometry average differs from averaging seed-level percentages or pooling objective values across differently scaled instances. Seeds describe variability conditional on an instance; they do not multiply the number of independent geometries. All scheduled configurations, ties, losses and failed runs remain visible. Best-seed selection is not used for the primary comparison.

The nine-case training screen requires a mean reduction of at least 0.2%, at least six positive geometry means, at most one negative mean, and no missing or invalid scheduled comparisons. These thresholds are an exploratory investment rule, not a significance test or an acceptance criterion for publication. The earlier validation screen retains its original, different threshold and its observed failure. It is not relabeled as successful because a later training experiment looks favorable.

A safety timeout of 43,200 seconds in an iteration-limited reference run means censoring, not completion. Without a completed reference, no valid $T_{g,s}$ is available for either corresponding candidate. Both candidate slots remain unavailable in the planned denominator. A last checkpoint is retained as an observation but does not replace the missing 70-iteration endpoint. No complete cost-only primary result is formed by dropping such pairs. Infrastructure interruptions are recorded separately from genuine algorithm or solver failures.

6.5 Deadlines, suspension and independent audits

Solution eligibility is determined from absolute monotonic event timestamps relative to the parent start time. A solution must be observed by the outer deadline and pass validation against the original instance. Cleanup or process return after that deadline does not create additional search time, nor does it erase an already eligible incumbent. Actual process durations, safety timeouts and cleanup overruns are retained separately. Descriptive curves at 1, 10, 30, 60, 120 and 300 seconds do not replace the primary 300-second endpoint.

The first longer-horizon collection was interrupted after host suspension was identified in the operating-system power log. Its 44 completed records are preserved with an explicit environmental-interruption manifest. They are not treated as a controlled time comparison. The replacement collection repeats all 81 scheduled executions in a separate directory, rather than combining apparently unaffected early cases with favorable later repetitions. It retains the same instances, seeds, parameters, method ordering and binary versions.

During the replacement collection, a monitor samples monotonic and wall clocks once per second. A monitoring gap above five seconds, a difference between clock increments above three seconds, or a recorded system-sleep event invalidates the block and stops the queue. Temporary idle-sleep prevention does not prevent sleep caused by closing the laptop lid. Each block records environmental diagnostics and hashes of its three results. A separate auditor checks those records. These operational checks screen for suspension and substantial scheduling interruptions; they do not prove identical CPU or thermal conditions. Equivalent protection is required before accepting the prospective iteration-limited comparisons.

One block in the replacement collection subsequently exceeded the monitoring limits without a recorded sleep event: its maximum monitoring gap was 9.408 seconds and its maximum clock-increment difference was 3.609 seconds. All three executions of that block were preserved as environmentally invalid and repeated together with the same methods,

seed, parameters and order. The other 23 completed valid blocks were retained. This decision used the recorded environmental thresholds, not the relative costs of the three methods. The completed replacement study contains 81 eligible executions; the three invalid attempts remain separately archived.

An independent scalar reader checks mandatory visits, unique locations, coverage, prize collection and the cycle cost against the original input matrix. Audits also reconstruct eligible incumbent selection and check input, source, binary and result hashes. Feasibility verification and a successful environment check address different questions; neither substitutes for the other. Historical finite verification records without source hashes are retained as historical evidence and cannot by themselves certify the final source revision. This partial manuscript makes no confirmatory significance claim or new best-known-solution claim from the exploratory comparisons.

7 Computational results

7.1 Completed exploratory comparisons

Effects use the equally weighted geometry means defined in Equation (13). Table 1 reports completed comparisons with their original denominators. Positive effects indicate lower candidate cost; repetitions within a geometry were not treated as independent benchmark instances.

Study	Runs	Reduction (%)	W/L/T
Charged TRAIN, 60 s	54	1.886	8/1/0
Conditional fixed-support replay	27	2.014	9/0/0
Limited validation, 60 s	70	0.907	2/1/4
Conditional joint insertion vs. greedy	324	-0.181	3/6/0

Table 1: Completed exploratory comparisons. Positive reduction favors the hybrid in the first three rows and nearest-gap joint insertion in the last. W/L/T counts geometry means. Conditional rows exclude generation of their common starting states and are not directly comparable with charged end-to-end rows.

The corrected 60-second training experiment comprised nine geometries, three seeds and two methods, for 54 executions. The hybrid reduced mean cost in eight geometry means and increased it in one, with an equally weighted reduction of 1.886%. Independent audits checked 334 routes, including intermediate incumbents. These observations followed several exploratory attempts and were not a confirmatory test of generalization.

In a conditional replay, fixed-support LKH received the remaining allowance from each of the 27 saved MDM initialization states. The hybrid had a lower seed-mean cost in all nine geometry groups, with a mean reduction of 2.014% relative to this control. The replay reused training starts and excluded their generation cost. It supplied a component comparison rather than an independent end-to-end replication or evidence for each destruction and repair decision separately.

Figure 3 displays the geometry-level heterogeneity behind the two training averages. The largest end-to-end reduction occurred on rd400, whereas kroA200 and ts225 showed small mean changes and tsp225 showed a small deterioration. Individual seed effects also

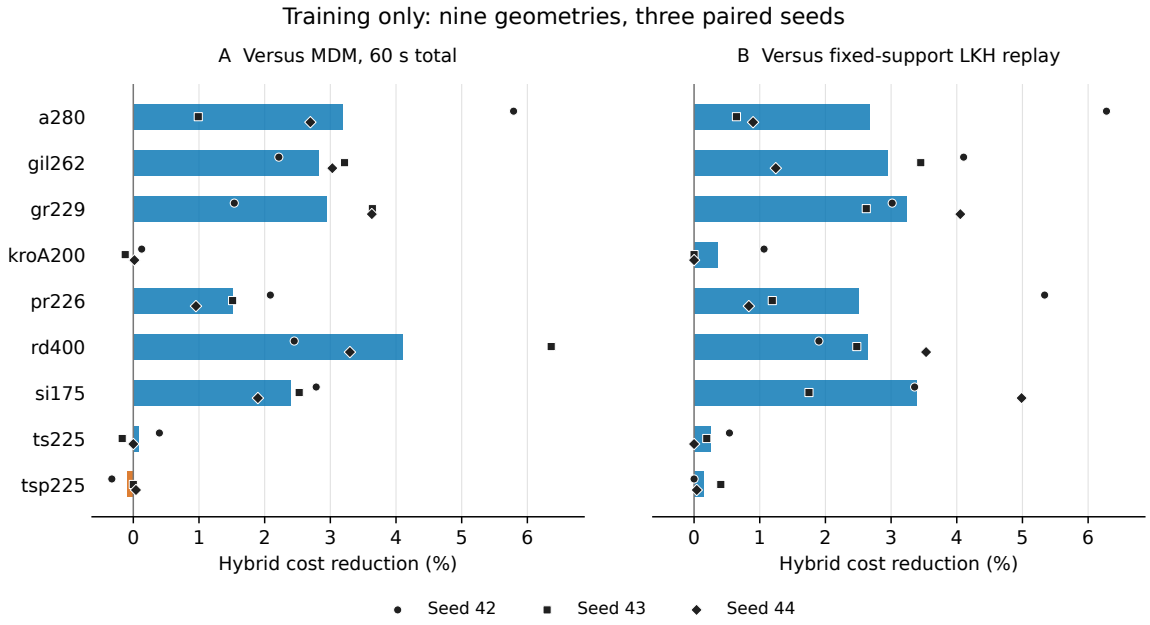


Figure 3: Completed exploratory training comparisons on nine geometries with three paired seeds. Bars show the relative reduction of seed-mean costs; markers show individual seed-pair reductions, not confidence intervals. Panel A compares fresh executions with MDM under a total 60-second budget, including initialization. Panel B uses a conditional fixed-support LKH replay from saved initialization states and excludes their generation cost. Positive values favor the hybrid. Neither panel provides independent test confirmation.

varied within geometries. The markers are paired observations, not uncertainty intervals; the two panels use different controls and cannot be combined into one effect.

The frozen validation configuration was evaluated on seven geometries with five seeds per method, yielding 70 executions. Audits checked 466 routes without missing runs or feasibility failures. The primary mean reduction was 0.907%, comprising two positive geometry means, four ties and one negative mean. The reductions on pr264 and pr299 were 2.730% and 3.771%, respectively; d198 increased by 0.150%. The prespecified screen required at least three positive geometry means, so the candidate did not pass. The twelve reserved test geometries remained unopened. These validation geometries had already been used for an earlier, different LKH configuration; the present screen therefore did not constitute untouched confirmation.

A separate conditional training experiment compared greedy repair with joint insertion MIP repair after matched destructions. Across 324 executions, nearest-gap MIP repair had a mean reduction of -0.181% relative to greedy repair and failed its prespecified mean-improvement screen. It improved on the equal-size random-gap MIP control in eight geometry means. All 1,296 audited routes were feasible, including retained fallback incumbents; this does not imply that every restricted repair model was feasible. The experiment excluded generation of common MDM starts and did not establish an end-to-end advantage for MIP integration.

Earlier clock-collector experiments are retained as separate negative records. Their wrapper could discard an improvement observed before the deadline when cleanup returned late. The corrected collection used absolute incumbent event times and fresh execu-

Geometry	MDM300	MDM10+repair	MDM60+repair	H10 (%)	H60 (%)
a280	2029.33	1998.00	2001.33	1.544	1.380
gil262	2016.33	1982.67	1985.00	1.670	1.554
gr229	98281.33	96722.33	96737.67	1.586	1.571
kroA200	23497.67	23505.33	23495.00	-0.033	0.011
pr226	68110.00	67286.00	67286.00	1.210	1.210
rd400	10773.67	12872.33	10670.33	-19.480	0.959
si175	7045.33	6908.00	6898.00	1.949	2.091
ts225	108998.00	108866.33	108866.33	0.121	0.121
tsp225	2447.00	2449.00	2449.00	-0.082	-0.082

Table 2: Completed 300-second TRAIN screen: costs averaged over seeds 70–72. H10 and H60 are geometry-level relative cost reductions against MDM300; positive values favor the hybrid. Both arms use 300 seconds total, including initialization. All nine geometries and both prespecified arms are retained.

tions; its differences from the earlier collection cannot be attributed solely to re-extracting identical trajectories. No result in this subsection establishes a new best-known solution, a literature-wide novelty claim or inferential significance after exploratory method selection. The completed longer-horizon screen is reported separately in Section 7.2 and is excluded from this table.

7.2 Completed 300-second training screen

The longer-horizon screen comprised 81 eligible executions: nine selected training geometries, three paired seeds and three methods. All 27 retained blocks passed the recorded environmental checks. The endpoint audit verified 624 routes, and the separate reconstruction of six time horizons checked 3,371 route-event records; these counts describe different audit passes and are not additional experimental repetitions. No scheduled final solution was missing or infeasible.

Table 2 reports the complete geometry means. The variant with up to 60 seconds of MDM initialization reduced the equally weighted mean cost by 0.979%, with eight positive geometry means and one negative mean. It passed the exploratory requirement of at least 0.2% mean reduction, at least six positive means and at most one negative mean. The variant with up to 10 seconds of initialization increased mean cost by 1.279%, with six positive and three negative geometry means, and failed the same requirement. Both alternatives and all nine geometries remain in the analysis.

Figure 4 shows the heterogeneity and the paired-seed observations on a shared scale. The largest deterioration occurred for the 10-second initialization on rd400: its seed-mean cost was 19.480% above MDM300. The 60-second variant instead reduced mean cost on that geometry by 0.959%. On tsp225, both hybrids returned 2449 in each seed, whereas MDM300 returned 2447, a deterioration of 0.082%. Small losses are retained alongside the larger effects rather than rounded into ties.

Passive mining exports also qualify the comparison. MDM300 recorded mining calls in 11 of its 27 executions; nine executions had a call provably within the deadline. The 60-second hybrid initialization recorded calls in two executions, one provably timely; the 10-second initialization recorded none. These are bounds derived from available cumulative exports, not estimates of the improvement caused by mining. The completed screen

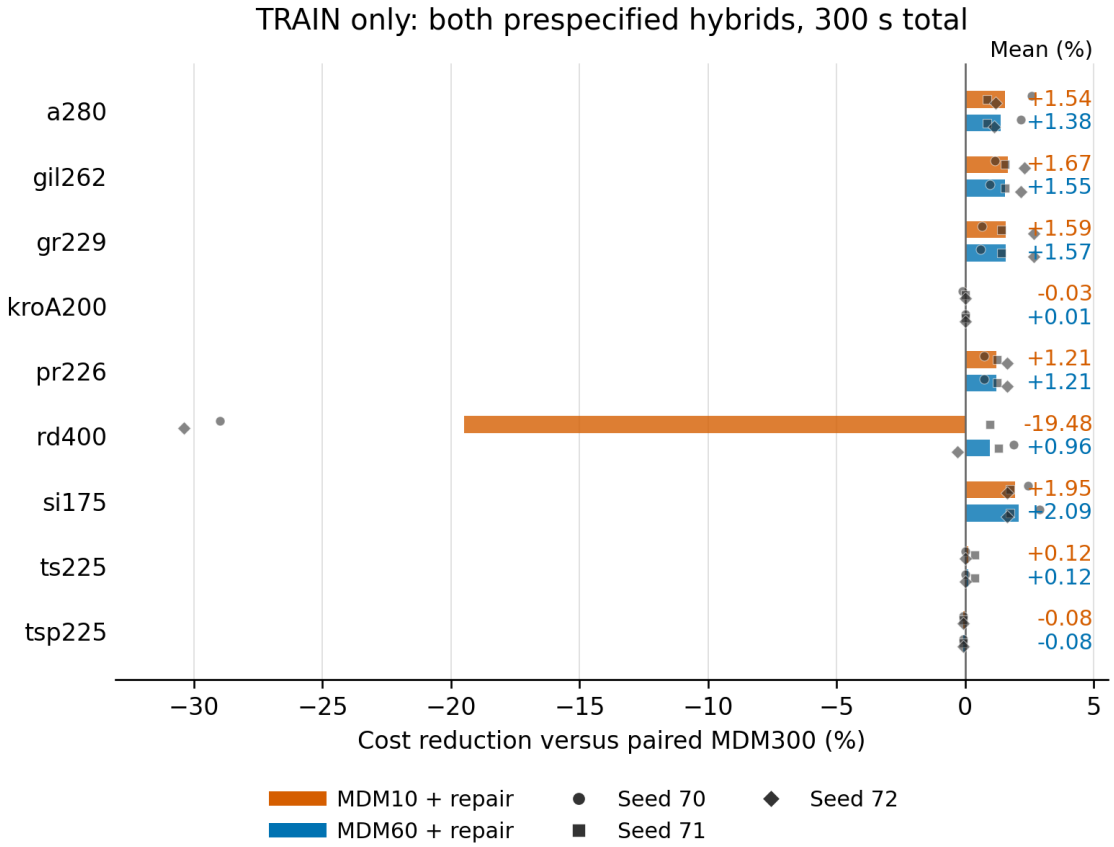


Figure 4: Completed 300-second screen on nine selected TRAIN geometries and three paired seeds. Bars show relative reductions of seed-mean costs; markers show individual seed-pair reductions, not confidence intervals. Both variants use the same horizontal axis and include initialization in the total budget. Positive values favor the hybrid. The right-hand labels give the geometry means. All losses are retained. This screen does not establish performance on reserved test geometries or superiority over a completed 70-iteration MDM reference.

therefore supports continued evaluation of the 60-second hybrid within the selected training set. It does not replace the iteration-limited reference comparison.

8 Statistical analysis and ablations

8.1 Geometry-level sensitivity of completed comparisons

The mean reduction alone does not describe how broadly improvements occur. We therefore reconstructed each geometry effect directly from the audited seed costs of the two completed fresh 60-second studies. Rational arithmetic was used to check the reported seed-mean effects before rounding. Training and reused validation were analyzed separately; their different geometry sets and selection histories preclude pooling them into one confirmation.

Study	n	Mean (%)	Median (%)	LOO mean range (%)
TRAIN	9	1.886	2.404	[1.609, 2.134]
Reused validation	7	0.907	0.000	[0.430, 1.084]

Table 3: Descriptive sensitivity of completed 60-second comparisons. The leave-one-geometry-out (LOO) range recomputes the equally weighted mean after each possible single omission. It is not a confidence interval or an alternative primary endpoint. All original geometries remain in the reported primary means.

For the nine training geometries, the median reduction was 2.404%, compared with a mean of 1.886%. Recomputing the mean after each possible single geometry omission gave values from 1.609% to 2.134% (Table 3). Thus, no single training geometry accounted for the entire positive average. This observation concerns the selected training set and does not establish performance on new geometries.

For the seven reused validation geometries, the median effect was zero, despite the positive mean of 0.907%. The single-omission means ranged from 0.430% to 1.084%. The breadth of improvement remained limited: four geometry means tied, one deteriorated and only two improved. As a retrospective concentration diagnostic, the five geometries other than pr264 and pr299 had a mean reduction of -0.030%. This is not an alternative primary comparison: both positive cases remain in the seven-case result. It explains why the positive overall mean can coexist with failure of the original requirement for at least three positive geometry means.

The omission ranges are descriptive sensitivity measures, not confidence intervals. Seed repetitions characterize stochastic variation conditional on each geometry; they are not additional independent benchmark geometries. Because several candidate methods had already been examined and validation geometries had been reused, no confirmatory hypothesis test is attached to these post hoc diagnostics. They neither change a screening threshold nor justify stopping the incomplete longer-horizon collection early.

8.2 Sensitivity of the completed 300-second screen

The same descriptive omission calculation was repeated for both prespecified 300-second variants, using exact rational reconstructions of the 81 audited endpoint costs. For the 60-second initialization, single-omission means ranged from 0.840% to 1.112%; no individual

geometry accounted for the positive average. This is a statement about the selected training set, not a confidence interval or evidence of generalization.

For the 10-second initialization, the range was -1.683% to 0.996%. The positive endpoint arose when rd400 was omitted, illustrating concentration of the observed deterioration. That geometry remains in the primary mean of -1.279% and the original screen still fails. Neither the omission nor the sensitivity range supplies a replacement selection rule. Paired seeds within one geometry were not treated as independent instances.

8.3 Passive ordering-call diagnostics

The completed hybrid records also contain cumulative counters for the ordering phase. We reconciled these counters across all 27 training runs and all 35 reused-validation runs, checking each result hash against the independent audit and the interpreted source code against its collection protocol. Table 4 reports the retained categories.

Study	Runs	Calls	Exit 0	Timeouts	Other
TRAIN	27	19,778	19,443	327	8
Reused validation	35	72,388	72,234	150	4

Table 4: Passive counters in completed 60-second hybrid runs. Calls are invocations of the ordering wrapper. Exit 0 records normal process exit; timeouts record a reported subprocess timeout; other combines nonzero or absent exit codes. These categories partition wrapper invocations, rather than validated improvements or feasible final solutions.

The counters require a distinction between a call to the ordering wrapper and a launched LKH subprocess. A wrapper call can exhaust its allowance during matrix preparation, yielding no process exit code. A zero exit code does not by itself show that an improving tour was returned, validated and accepted before the deadline. Similarly, the other/absent-exit category cannot distinguish a process failure from a preparation-only invocation. The retained aggregate records do not supply per-call output-validity or improvement counts. Final route feasibility and incumbent eligibility are checked separately.

In training, rd400 accounted for 277 of the 327 reported subprocess timeouts. Its 735 wrapper invocations therefore had a timeout fraction of 37.7%. The corresponding fraction across reused validation was 150 of 72,388 invocations, or 0.21%. Each of the four tied validation geometries had at most four such timeouts. Widespread timeouts thus do not describe the execution status of those tied cases. This observation does not explain their objective values or establish optimality of their visited supports. The counters can include a final call returning after the outer deadline; they are not counts restricted to eligible improvement events. No operator, budget allocation or screening threshold is changed by this diagnostic.

9 Discussion and limitations

9.1 Visit changes, coverage and prize restoration

The fixed-support replay provides conditional evidence that reordering alone does not account for all of the observed training advantage. Both alternatives started from the

same saved MDM states and had the corresponding remaining allowance. The comparison nevertheless excludes generation of those states, uses previously examined training geometries and evaluates the complete visit-changing phase. It cannot separately identify the effect of the destruction size, greedy scoring, temporary infeasibility, acceptance rule or sequence of LKH calls.

A descriptive check of the 27 completed training trajectories found that the final visited set differed from the pre-repair set in 23 runs. Removing the locations absent from the final set, without yet adding the final new locations, would violate the prize quota in 20 runs and coverage in five. The five coverage cases were included among the 20 quota cases. These are feasibility witnesses for net differences between two feasible endpoints. They do not reconstruct an intermediate destruction state, establish the necessity of a particular insertion, or prove which search component caused the final improvement.

The input structure further limits a coverage-based explanation. In gr229, kroA200, pr226 and rd400, the mandatory locations already cover every demand. Because all admissible tours retain those locations, each coverage constraint is automatically satisfied in these four instances. Optional-visit changes there still affect routing and prize collection, but their gains cannot be attributed to restoring missing demand coverage. In all nine selected training instances, mandatory prize alone is insufficient. Thus the available evidence supports examining prize and visit selection alongside coverage, rather than assuming that coverage restoration is the dominant mechanism throughout the training set.

9.2 Restricted integer repair and computational allocation

The evaluated joint-insertion model did not improve on greedy repair in the primary mean comparison. Its advantage over the equal-size random-gap control suggests that the chosen insertion positions matter within that experiment; it does not establish an advantage for MIP repair over the simpler repair routine. Restricting candidates and positions can exclude useful tours, and a model that restores coverage and prize exactly need not find a lower-cost tour within its allowance. The present comparison does not isolate which restriction or runtime component explains the unfavorable mean effect.

The same distinction applies to path contraction and local optimality. An exact representation of a restricted repair preserves the meaning of its feasible solutions and objective values. It does not show that solving the restricted model is a good allocation of an end-to-end search budget. Likewise, an optimal restricted repair is not a certificate for the unrestricted PCCTP. The negative repair result is retained separately from the uniform-removal, greedy-repair hybrid; the successful training screen of the latter cannot be used as evidence for an integer component that it did not execute.

9.3 Limits of the current performance evidence

The training set contains nine selected difficult geometries, rather than a representative sample of the full published evaluation set. The seven validation geometries have also been reused. Their two positive means and zero median, together with the failed breadth criterion, limit any claim of general improvement. The twelve reserved test geometries remain unopened; their existence is not evidence of performance on them. Descriptive omission analyses do not repair selection bias or supply independent confirmation.

Search horizons create another limitation. A short-budget comparison can end before

MDM activates its mining component. It then compares the observed algorithm trajectories within that horizon, rather than isolating the value of mining or outperforming a completed 70-iteration reference. The completed 300-second experiment favored the 60-second initialization variant in the selected training set, while the 10-second variant failed its screen. This difference cautions against treating additional repair time as a substitute for adequate initialization. It does not isolate the causal effect of initialization length, because the resulting search trajectories and computational allocation also differ. The planned iteration-limited comparisons address a different stopping rule and remain necessary. The reference-defined duration rule also requires disclosure of the reference-first order and possible machine-state drift within a pair.

Finally, the recovered baseline includes declared decoder and elite-count corrections, and measured times refer to the present hardware and executable versions. They do not establish speedups over seconds reported on the older processor. Any final performance contribution requires valid complete comparisons, a separately frozen broader evaluation and an account of failures, mining exposure and computational cost. The completed results currently justify continued evaluation of the visit-changing hybrid, while retaining the failed validation screen and negative integer-repair comparison as constraints on its interpretation.

10 Conclusions

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